

A Range-Core Obstruction and an Adjoint Weighted-Shift Dichotomy for Quasinilpotent Differential-Algebraic Equations

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Status: substantial partial answer, likely valid. Every solution of $(Tx)' = x$ takes its values in $\mathcal{H}_c(T) := \bigcap_{n \geq 1} \overline{\text{ran } T^n}$ and generates a nonzero closed invariant subspace on which T has dense range. Consequently, a total generalized kernel of T^* forces triviality. For the compact quasinilpotent forward shift $Fe_n = (n+1)^{-1}e_{n+1}$ the DAE is trivial, whereas its adjoint has an infinite-dimensional family of explicit global solutions. Thus compactness, spectrum, singular values, and even all power norms do not determine the answer.

1 Source problem

Friedrich M. Philipp's *On Differential-Algebraic Equations with Bounded Spectrum in Banach Spaces*, arXiv:2504.10346v1, calls $x \in C([0, \infty), \mathcal{X})$ a solution of

$$\frac{d}{dt}Tx = x \tag{1}$$

when $Tx \in C^1([0, \infty), \mathcal{X})$ and $(Tx)' = x$. Open Problem 4.3 asks for which quasinilpotent $T \in \mathcal{L}(\mathcal{X})$ equation (1) has only the zero solution [1].

$[0, \tau]$ and is therefore given by $z(t) = T_t z(0)$. Hence, $x(t) = T_{\tau-t}x(\tau) = 0$ for $t < \tau - 1$. Since $\tau > 1$ was arbitrary, the claim follows. We further would like to point out that the function $x(t) = T_{\tau-t}x(\tau)$ solves the IVP $\frac{d}{dt}Tx = x$, $x(0) = 0$, on $[0, \tau]$, but does not vanish.

In view of Example 4.2, the question arises for which quasi-nilpotent operators the DAE (4.1) only has the trivial solution on $[0, \infty)$. We pose this as an open problem.

Open Problem 4.3. For which quasi-nilpotent operators $T \in \mathcal{L}(\mathcal{X})$ does the DAE $\frac{d}{dt}Tx = x$ only have the trivial solution $x = 0$ on $[0, \infty)$?

Figure 1: Open Problem 4.3 in arXiv:2504.10346v1, PDF page 11.

2 Proof intuition

The equation supplies much more regularity after applying powers of T : $T^n x$ is n times continuously differentiable and its n th derivative is x . Since a derivative is a limit of finite differences,

and all finite differences of $T^n x$ lie in $\text{ran } T^n$, every value $x(t)$ lies in the closed range of every power. Equivalently, $x(t)$ is annihilated by every functional killed by a power of T^* .

The discrete shift examples expose the strength and the limitation of this obstruction. A forward shift successively forces the coordinates of a solution to vanish. Its adjoint is a backward shift; prescribing the first coordinate then turns the remaining coordinates into successive terminal antiderivatives. Fast-decaying weights make these antiderivatives square-summable for a short compactly supported seed.

3 The range-core obstruction

The following result does not require quasinilpotence.

Theorem 1 (closed-hyperrange and invariant-core obstruction). *Let $T \in \mathcal{L}(\mathcal{X})$ on a complex Banach space, and let x solve (1). Then, for every $n \geq 1$,*

$$T^n x \in C^n([0, \infty), \mathcal{X}), \quad \frac{d^k}{dt^k} T^n x = T^{n-k} x \quad (0 \leq k \leq n). \quad (2)$$

In particular,

$$x(t) \in \mathcal{H}_c(T) := \bigcap_{n \geq 1} \overline{\text{ran } T^n} \quad (t \geq 0). \quad (3)$$

If $x \neq 0$, then

$$M := \overline{\text{span}}\{T^k x(t) : k \geq 0, t \geq 0\} \quad (4)$$

is a nonzero closed T -invariant subspace satisfying

$$M = \overline{T(M)}. \quad (5)$$

Consequently, either of the following conditions forces every solution to be zero:

1. $\mathcal{H}_c(T) = \{0\}$;
2. $\bigcup_{n \geq 1} \ker(T^*)^n$ is total in $\mathcal{X}^*$;
3. no nonzero closed T -invariant subspace M satisfies $M = \overline{T(M)}$.

Proof. The assertion for $n = 1$ is the definition of a solution. Suppose (2) holds for n . Applying T preserves all existing derivatives, and

$$\frac{d^n}{dt^n} T^{n+1} x = T x \in C^1, \quad \frac{d^{n+1}}{dt^{n+1}} T^{n+1} x = x.$$

This proves (2) by induction.

For fixed $t \geq 0$, the forward n th finite-difference quotients of the C^n function $T^n x$ converge to $(T^n x)^{(n)}(t) = x(t)$. Each quotient is a linear combination of values $T^n x(t + jh)$ and hence belongs to $\text{ran } T^n$. This proves (3), including at $t = 0$.

The space in (4) is nonzero when $x \neq 0$, is closed by definition, and is T -invariant. Each generator is a derivative,

$$T^k x(t) = \frac{d}{dt} T^{k+1} x(t),$$

and therefore is a limit of difference quotients of values $T^{k+1} x(s) = T(T^k x(s)) \in T(M)$. Thus $M \subseteq \overline{T(M)}$; the opposite inclusion follows from invariance.

Condition 1 now follows from (3), and condition 3 from (5). Finally, the standard annihilator identity

$$\overline{\text{ran } T^n} = {}^\perp \ker(T^*)^n$$

shows that every $x(t)$ annihilates $\bigcup_n \ker(T^*)^n$. Totality gives $x(t) = 0$ for every t . $\square$

4 Adjoint compact shifts with opposite behavior

Let $H = \ell^2(\mathbb{N}_0)$ with standard basis $(e_n)_{n \geq 0}$, and define

$$Fe_n = \frac{1}{n+1}e_{n+1}, \quad B = F^*. \quad (6)$$

Thus $Be_0 = 0$ and $Be_{n+1} = (n+1)^{-1}e_n$.

Theorem 2 (adjoint dichotomy). *The operators F and B are compact, nonnilpotent, and quasinilpotent, and*

$$\|F^m\| = \|B^m\| = \frac{1}{m!} \quad (m \geq 1). \quad (7)$$

They nevertheless have opposite behavior for (1):

1. *for $T = F$, the only solution is $x = 0$;*
2. *for $T = B$, there is an infinite-dimensional family of nonzero global solutions.*

More explicitly, choose $0 < a < 1$ and $f \in C_c([0, \infty))$ with $\text{supp } f \subseteq [0, a]$. Put

$$x_0(t) = f(t), \quad x_n(t) = (-1)^n n \int_t^a (s-t)^{n-1} f(s) ds \quad (n \geq 1), \quad (8)$$

where the integral is declared zero for $t \geq a$. Then $x \in C([0, \infty), H)$, $Bx \in C^1([0, \infty), H)$, and $(Bx)' = x$. The map $f \mapsto x$ is injective.

Proof. The weights of F tend to zero, so F , and hence $B = F^*$, is compact. For $m \geq 1$,

$$\|F^m\| = \sup_{j \geq 0} \frac{1}{(j+1)(j+2) \cdots (j+m)} = \frac{1}{m!}.$$

The same equality holds for $B^m = (F^m)^*$. Formula (7) gives spectral radius zero, while neither shift is nilpotent.

Suppose first that $(Fx)' = x$. The zeroth coordinate of Fx is identically zero, hence $x_0 = 0$. Since $(Fx)_{n+1} = x_n/(n+1)$, the equation gives $x_{n+1} = x'_n/(n+1)$. Induction yields $x_n = 0$ for every n .

For B , formula (8) gives

$$x'_{n+1}(t) = (n+1)x_n(t) \quad (n \geq 0). \quad (9)$$

At the endpoint a this remains valid because a continuous f supported in $[0, a]$ has $f(a) = 0$. Moreover,

$$|x_n(t)| \leq \|f\|_\infty (a-t)^n \leq \|f\|_\infty a^n \quad (0 \leq t \leq a),$$

and all coordinates vanish for $t \geq a$. The geometric majorant gives an ℓ^2 tail uniformly in t , so coordinatewise continuity upgrades to H -valued continuity.

Now $(Bx)_n = x_{n+1}/(n+1)$. Hence (9) gives the desired derivative coordinatewise. To verify the Banach-valued derivative rather than merely its coordinates, for $0 \leq u \leq t$ we use (9) to obtain

$$Bx(t) - Bx(u) = \int_u^t x(s) ds$$

in every coordinate. Both sides belong to H , so the equality holds in H . Continuity of x now implies $Bx \in C^1$ and $(Bx)' = x$. Finally, $x_0 = f$, so the construction is injective; the seed space is infinite-dimensional. $\square$

For orientation, the obstruction of Theorem 1 detects the same dichotomy:

$$\overline{\text{ran } F^n} = \{y : y_0 = \cdots = y_{n-1} = 0\}, \quad \mathcal{H}_c(F) = \{0\},$$

whereas $\text{ran } B^n$ contains every finitely supported sequence and therefore $\mathcal{H}_c(B) = H$.

5 A weighted-shift sufficient condition for nontriviality

The construction is not tied to the particular weights above.

Proposition 1. *Let B_w be the bounded backward weighted shift on $\ell^2(\mathbb{N}_0)$,*

$$B_w e_0 = 0, \quad B_w e_n = w_n e_{n-1} \quad (n \geq 1),$$

where every $w_n \neq 0$, and put $W_n = w_1 \cdots w_n$. If for some $a > 0$

$$\left(\frac{a^n}{n! |W_n|} \right)_{n \geq 1} \in \ell^2, \tag{10}$$

then $(B_w x)' = x$ has an infinite-dimensional family of compactly supported global solutions. If also $w_n \rightarrow 0$, then B_w is compact and quasinilpotent.

Proof. For a seed $f \in C_c([0, \infty))$ supported in $[0, a]$, set $x_0 = f$ and

$$x_n(t) = \frac{(-1)^n}{W_n(n-1)!} \int_t^a (s-t)^{n-1} f(s) ds \quad (n \geq 1).$$

Then $w_{n+1}x'_{n+1} = x_n$ and

$$|x_n(t)| \leq \|f\|_\infty \frac{a^n}{n! |W_n|}.$$

Condition (10) supplies the uniform ℓ^2 majorant, and the proof of Theorem 2 applies verbatim. If $w_n \rightarrow 0$, finite-rank truncations converge to B_w in norm, so B_w is compact. To see quasinilpotence directly, fix $\varepsilon > 0$ and choose N so that $|w_n| \leq \varepsilon$ for $n > N$. With $C = \max(1, \sup_n |w_n|)$, every product of $m > N$ consecutive weights is at most $C^N \varepsilon^{m-N}$ in modulus. Hence $\limsup_m \|B_w^m\|^{1/m} \leq \varepsilon$, and the spectral radius is zero. $\square$

For $w_n = 1/n$, one has $W_n = 1/n!$, and (10) reduces to $(a^n) \in \ell^2$, explaining the restriction $a < 1$ in Theorem 2.

6 Scope, verification, and novelty bound

Theorem 1 is a checkable sufficient condition, not a full characterization. Indeed, the source proves that the Volterra operator V has only the zero global solution even though V has dense range in $L^2(0, 1)$, so $\mathcal{H}_c(V) = L^2(0, 1)$. Thus a nonzero closed hyperrange or a dense-range invariant core does not by itself produce a solution.

The adjoint pair shows that the answer is not determined by compactness, quasinilpotence, singular values, or the sequence $(\|T^m\|)_{m \geq 1}$: F and F^* share all of these data. It also supplies a discrete analogue of the source’s sign-sensitive Volterra examples, with every solution written explicitly.

The proofs were checked against the solution definition in the current arXiv v1. A bounded novelty search through 21 July 2026 covered the run registry, the local arXiv source corpus, the source paper and its references, arXiv, and exact web combinations of “quasinilpotent”, “weighted shift”, “ $(Tx)' = x$ ”, “closed hyperrange”, and generalized kernels of T^* . It found the source paper and general weighted-shift literature, but no prior statement of Theorem 1, the explicit adjoint dichotomy, or Proposition 1 in this DAE setting. Publication-level novelty remains subject to expert review. We recommend review as a candidate partial result, with particular attention to the novelty of the closed-range criterion and the weighted-shift family rather than to their elementary ingredients.

References

- [1] F. M. Philipp, *On Differential-Algebraic Equations with Bounded Spectrum in Banach Spaces*, arXiv:2504.10346v1 (2025).